Aarav Jain

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EDUCATION

Purdue University | Purdue Engineering — **West Lafayette, IN**
Candidate for Bachelor of Science in Computer Engineering — Expected *May 2028*

SKILLS

Languages: Python, HTML/CSS/JavaScript, TypeScript, C/C++, Java, SQL, Matlab
Technologies: IsaacSim, Zephyr, PyTorch, ROS, React, React Native, Docker, AWS, Git, NextJS, Langchain, NumPy, FastAPI,

WORK & LEADERSHIP EXPERIENCE

Assured Guaranty — New York, NY
Software Engineering Intern — Feb. 2026 – Aug. 2026

- Built an automated credit memo generation pipeline using AWS Textract, Bedrock, and Snowflake by developing novel extraction and aggregation methods across multiple financial documents reducing analyst time per credit by 70%.
- Developed a CUSIP similarity engine using the weighted K-Nearest Neighbors algorithm and LightGBM leaf embeddings

Embedded Systems — West Lafayette, IN
Member — Sep. 2025 – Present

- Developing a Hardware-in-the-Loop (HiL) with multi-threaded architecture managing flight control, landing, autonomous hover, and controlled flight threads that process simulated sensor data to validate drone firmware in a safe environment
- Engineering device drivers in Zephyr RTOS for **IMU and 1D LiDAR sensors using I2C** communication protocols, implementing low-level hardware interfaces to enable real-time sensor data simulation and firmware validation

Purdue IDEAS Research Lab
Undergraduate Researcher — Oct. 2025 – Present

- Developing an autonomous UAV-UGV natural language collaboration system using LLMs, VLMS, and computer vision to enable navigation and goal-oriented behavior in unstructured environments, leveraging terrain for strategic movement
- Creating a simulation perception pipeline in Unity that captures UGV position and imagery to feed a **PyTorch-based VLM** perception system, generating a structured 3D object map for semantic understanding of unstructured environments
- Implementing intelligent control algorithms using reinforcement learning to improve UAV-UGV communication

Algoverse
Machine Learning Researcher — Aug. 2024 – May. 2025

- Developed and deployed a novel evaluation framework using Python and statistical analysis to quantify sycophantic behavior in multi-turn LLM conversations, identifying a **47% accuracy decline** over extended interactions
- Co-authored a research paper documenting the "Truth Decay" methodology, experimental pipeline, findings, and best-practice recommendations; **Published at the NAACL 2025 Student Research Workshop (SRW)**

PROJECTS

Datasheet Parser | Python, Typescript, NextJS, Pinecone, Modal, Docling. — github.com/Aarav-J/datasheet_parse

- Built an **electrical component datasheet intelligence platform** using Next.js, React, OpenAI, and Pinecone enabling engineers to ask natural-language questions over datasheets through **citation-backed Retrieval Augmented Generation**
- Developed a custom datasheet parsing pipeline with **Modal, Docling, S3, and LLM enrichment** to convert complex datasheets into structured data like electrical specifications, pin tables, register maps, and communication protocols
- Designed and implemented an engineering workspace for hardware components with AI explanations, multi-PDF component uploads, comparison tools, annotations, saved notes, and project boards for design decisions

AaRTOS | C, ARM Assembly, STM32F4, libopencm3 — http://github.com/Aarav-J/aaRTOS

- Built a preemptive **RTOS kernel for ARM Cortex-M4** with PendSV-driven context switching, SVC-based task startup, 1ms SysTick preemption, and round-robin scheduling to execute concurrent tasks on bare-metal STM32F4 hardware
- Implemented **thread synchronization** including priority-aware blocking mutexes and **inter-task message queues** to enable safe resource sharing, task communication, and deterministic coordination across concurrent execution contexts
- Developed a bare-metal firmware validation framework with LED blinking and UART printing tasks sharing a mutex-protected serial interface to verify preemptive scheduling, context-switch integrity, and synchronization correctness.